Chapter 4 exercises

Use real Euclidean inner products unless stated otherwise. Specify the projected subspace, basis, rank assumptions, and residual.

CH04-E1

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

1. Find a basis of $N(A)$.
2. Verify that a row-space basis is orthogonal to the nullspace.
3. Find $N(A^T)$ and explain why $C(A) = \mathbb{R}^2$.

CH04-E2

Let $a = (1, 1, 1)^T$ and $b = (2, 1, 4)^T$.

1. Find the projection p of b onto $L = \operatorname{span}\{a\}$.
2. Find $e = b - p$.
3. Check $a^T e = 0$.
4. Give P and verify $P = P^T = P^2$.

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CH04-E3

Fit $y = c + mt$ to the points

$$(t, y) = (0, 1), (1, 2), (2, 2).$$

1. Give the design matrix A and parameter vector x.
2. Form and solve the normal equations.
3. Find p, e and check $A^T e = 0$.
4. Find the minimum sum of squared residuals.

CH04-E4

Let $A \in \mathbb{R}^{m \times n}$ have independent columns and define

$$P = A(A^T A)^{-1} A^T.$$

Chapter 4 exercises (continued)

1. Prove $P^T = P$ and $P^2 = P$.
2. Prove $I - P$ is a projector and $P(I - P) = 0$.
3. Prove $A^T A$ is invertible.
4. Conversely, prove dependent columns make $A^T A$ singular.

CH04-E5

For

$$a_1 = (1, 1, 0)^T, \qquad a_2 = (1, 0, 1)^T$$

perform Gram-Schmidt.

1. Find orthonormal q_1, q_2.
2. Give R in $A = [a_1\ a_2] = QR$.
3. Check $Q^T Q = I_2$ and $QR = A$.

CH04-E6

Use A from CH04-E5 to solve $Ax = b$ by least squares, with

$$b = (1, 2, 0)^T.$$

1. Compute $Q^T b$ explicitly and solve $Rx = Q^T b$.
2. Find $p = Ax$, $e = b - p$, and $\|e\|^2$.
3. Check $A^T e = 0$ column by column without first assuming the normal equations.

CH04-E7

Let

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

1. Determine whether the columns are independent.
2. Project b onto $C(A)$.
3. Find every least-squares solution $x^\star$.
4. Explain why projection is unique but parameters are not.

CH04-E8

Let

Chapter 4 exercises (continued)

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

1. Prove $P^2 = P$.
2. Prove $P \neq P^T$.
3. Find the range and kernel of P.
4. Explain why idempotence does not imply orthogonal projection, identifying the failure of orthogonality between the range and kernel.

CH04-E9

For $A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \end{bmatrix}$ and $b = (3, 1)^T$, find bases and dimensions for all four fundamental subspaces. Compute the orthogonal projection p and residual e. Prove the minimum residual norm using a Pythagorean identity and find all minimizing parameters x. Prove which parameter has minimum norm. Finally decompose the particular solution $(2, -1, 2)^T$ into row-space and nullspace parts and compare its norm with that minimum.

CH04-E10

In exact arithmetic, fit $y = c + mt$ to $(0, 2), (1, 0), (2, 1)$.

1. Write the design matrix and normal equations. Show elimination steps for the intercept and slope.
2. Tabulate predictions, residuals and their squares; check $A^T e = 0$ and draw the vertical residuals in the observation plot.
3. Derive the exact increase in squared residual norm under an arbitrary parameter change $(\Delta c, \Delta m)$. Prove uniqueness. Why is a zero residual sum insufficient?
4. Center time using $s = t - 1$, then independently check the parameters and minimum residual with thin QR and $q_3 = (1, -2, 1)^T/\sqrt{6}$. Distinguish vertical direction in the observation plot from residual orthogonality in $\mathbb{R}^3$.

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References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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